

Shujaullah Ahsan

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Boston, MA

Education

Northeastern University, Boston, MA September 2024 – Present
Masters in Data Science (Relevant Coursework: Introduction to Data Management and Processing, Web Development.)

University of Massachusetts Boston, Boston, MA September 2019 – May 2022
Bachelor of Science in Computer Science (Relevant Coursework: Data Structures and Algorithms, Higher Level Languages, Java Compiler, Operating Systems, Database Management, Android Development, Theory of Computation.)

Quincy College, Quincy, MA September 2017 – May 2019
Associate in Computer Science (Relevant Coursework: Advanced Java Programming, Data Structures and Algorithms in C++, Linux Administration, Robotics.)

Skills and Technologies

Programming: Python, Java, JavaScript, C++, C#, Bash

Databases: MySQL, PostgreSQL

Frameworks & Libraries: Django, React, Spring Boot, Qt, .NET

Embedded & Tooling: RS-232 Serial, ADB, U-Boot, Arduino/firmware, RFID

Backend & Cloud: AWS (S3), Heroku

Tools: Git, GitHub, Bitbucket, IntelliJ, VS Code, Android Studio, PyCharm, Visual Studio

Operating Systems: Windows, Linux, macOS

Familiar With: R, MongoDB, JSON, AI, TypeScript, Racket

Work Experience

Software Development Engineer, Medica Corporation, Boston, MA November 2022 - Present

- Designed and implemented the **EasyLyte** QC testing application in C++/Qt 6, improving measurement accuracy and reducing manual effort in laboratory workflows.
- Engineered high-performance multithreaded C++ modules for real-time data acquisition and processing, significantly enhancing device responsiveness.
- Optimized RS-232 serial communication protocols and data visualization pipelines, enabling reliable interaction with medical analyzers in production environments.
- Implemented two-way communication in Java between the Android-based analyzer and the desktop application, parsing and validating real-time instrument data streams passed from the device into the software.
- Developed Arduino firmware for a custom test fixture to drive and exercise instrument hardware during manufacturing and QA.
- Built automated board-provisioning tooling that scripts **U-Boot** over serial and **ADB** to flash the OS image and deploy APKs, reducing manual setup time on the production line.
- Integrated MySQL for persistent storage of calibration and QC data; developed SQL-driven reporting tools that provided manufacturing teams with actionable insights.
- Developed RFID programming solutions using C#/.NET, streamlining sample tracking and instrument authentication processes.
- Partnered with cross-functional teams (QA, manufacturing, product management) under Agile/Scrum, delivering production-ready software in iterative sprints.

UNIX Lab Operator, University of Massachusetts Boston January 2020 – May 2022

- Developed and deployed a CS registration portal using Django, HTML, CSS, and JavaScript, streamlining student enrollment processes.
- Built secure authentication workflows and intuitive dashboards, improving user experience for students and faculty.
- Automated system monitoring and maintenance with shell scripts and cron jobs, reducing downtime and enhancing lab reliability.
- Supported faculty and students in resolving UNIX system issues, improving lab productivity and system uptime.

Projects

Matched (Django, AWS S3, HTML, CSS, JavaScript)

Led a team to develop a post-retirement career portal for pro athletes; implemented user registration, authentication, dashboard features; deployed on Heroku.

JDBC Shop (JDBC, Oracle SQL)

Created a command-line shop management system with customers, products, and sales modules using Oracle backend.

Compiler Project (Java)

Developed a subset Java compiler capable of token validation, parse tree generation, syntax checking, and bytecode output generation.

Achievements

- 2022 Departmental Distinction, UMASS Boston CS Department
- 2021 & 2022 Dean's List, UMASS Boston
- 2021 & 2022 UMASS Boston Computer Science Scholarship
- 2018 Quincy College Codeathon Winner

Additional Information

Adaptable to new technologies and programming paradigms; keen interest in Data Science and Machine Learning; active problem-solver on LeetCode.